Future Interns – Cyber Security Task 2 (2026)

Phishing Email Detection & Awareness Report

Submitted By: ___

Intern ID: ___

Date: ___

Organization: Future Interns

---

1. Objective

The objective of this task is to analyze phishing emails, identify phishing indicators, classify email risk levels, and create awareness guidelines to help users recognize and avoid phishing attacks.

---

2. Tools Used

Tool	Purpose
Google Message Header Analyzer	Email header analysis
MXToolbox Email Header Analyzer	Header investigation
Web Browser	URL inspection
GitHub Phishing Repositories	Sample phishing emails
MS Word / Google Docs	Documentation

---

3. Sample Phishing Email Collected

Email Sample 1

Subject:

⚠️ Urgent: Your Account Will Be Locked

Sender:

security@secure-account-verify.com

Email Body:

Dear User,

We noticed suspicious activity on your account.

To avoid account suspension, please verify your details immediately.

Verify Now: http://secure-account-verify.com

Failure to verify within 24 hours will result in permanent account lock.

Regards,

Security Team

Screenshot Evidence

[Insert Screenshot 1 – Full Email Sample Here]

---

4. Email Header Analysis

The email header was analyzed using online header analysis tools.

Findings

- Sender domain appears suspicious.
- SPF validation failed.
- Domain does not belong to any legitimate organization.
- Return-Path differs from displayed sender address.
- Possible spoofing indicators detected.

Screenshot Evidence

[Insert Screenshot 2 – Email Header Analysis Result]

---

5. Link and Domain Inspection

Suspicious URL

http://secure-account-verify.com

Observations

- Domain name is not associated with a trusted company.
- Uses fear and urgency tactics.
- URL attempts to mimic a legitimate security portal.
- Users may be redirected to a fake login page.

Screenshot Evidence

[Insert Screenshot 3 – URL Inspection Result]

---

6. Phishing Indicators Identified

The following phishing indicators were detected:

Indicator	Found
Suspicious Sender Address	Yes
Generic Greeting	Yes
Urgency/Fear Language	Yes
Suspicious URL	Yes
Request for Sensitive Information	Yes
Brand Impersonation	Possible

Explanation

The email creates panic by claiming the account will be locked within 24 hours. The sender urges immediate action through a suspicious link, increasing the likelihood that users will disclose credentials.

Screenshot Evidence

[Insert Screenshot 4 – Highlighted Phishing Indicators]

---

7. Risk Classification

Classification: PHISHING

Risk Level: HIGH

Reason

The email contains multiple phishing characteristics:

- Fake sender domain
- Suspicious verification link
- Urgency-based social engineering
- Generic greeting
- Credential harvesting attempt

Therefore, the email should be classified as a phishing attack.

Screenshot Evidence

[Insert Screenshot 5 – Risk Classification]

---

8. How the Attack Works

Step 1

Attacker sends a fraudulent email.

Step 2

Victim receives warning message.

Step 3

Victim clicks malicious link.

Step 4

Victim enters login credentials.

Step 5

Attacker steals credentials.

Step 6

Attacker gains unauthorized access.

Screenshot Evidence

[Insert Screenshot 6 – Attack Flow Diagram]

---

9. Prevention Guidelines

For Employees

Do

✔ Verify sender addresses carefully.

✔ Check domain names before clicking links.

✔ Report suspicious emails to the security team.

✔ Enable Multi-Factor Authentication (MFA).

✔ Hover over links before clicking.

✔ Verify urgent requests through official channels.

Don't

✘ Click suspicious links.

✘ Download unexpected attachments.

✘ Share passwords or OTPs.

✘ Trust emails creating panic or urgency.

✘ Provide sensitive information through email.

---

10. User Awareness Tips

1. Legitimate companies rarely ask for passwords via email.
2. Check for spelling mistakes and unusual domains.

3. Be cautious of messages demanding immediate action.
4. Verify requests through official websites.
5. Report suspicious emails immediately.

---

11. Conclusion

The analyzed email demonstrates several common phishing techniques including urgency, impersonation, suspicious links, and credential theft attempts. Users must be trained to recognize these indicators and follow security best practices to prevent successful phishing attacks.

Proper awareness and verification procedures significantly reduce the risk of phishing-related incidents.

References

1. Google Message Header Analyzer
2. MXToolbox Email Header Analyzer
3. Future Interns Task Guidelines
4. Public Phishing Email Sample Repositories

Screenshots Checklist

☐ Screenshot 1 – Sample Phishing Email

☐ Screenshot 2 – Header Analysis

☐ Screenshot 3 – URL Inspection

☐ Screenshot 4 – Phishing Indicators Highlighted

☐ Screenshot 5 – Risk Classification

☐ Screenshot 6 – Attack Flow Diagram

End of Report